Assignment 7

Sample Data Insertion

Realistic seed records for all six LibSphere tables

Project: LibSphere - Library Management System **Prepared:** 23 October 2026

Platform assumption: MySQL 8.0.16 or later (InnoDB). Outputs in this report are expected results derived from the included sample dataset; run the SQL in MySQL Workbench to capture authentic execution screenshots if required by the instructor.

The sample dataset contains five categories, nine authors, eight book titles, nine book-author links, six members, and fourteen borrowing transactions. It covers one co-authored book, active and suspended memberships, currently issued/overdue loans, and returned loans. Contact details use example.edu addresses and illustrative phone values.

1. Recommended insertion order

Insert parent rows first, then bridge and transaction rows: Categories and Authors -> Books -> Book_Authors -> Members -> Loans. This order satisfies all foreign-key dependencies.

2. Insert categories and authors

```
INSERT INTO Categories (category_id, category_name) VALUES
(1, 'Computer Science'),
(2, 'Fiction'),
(3, 'History'),
(4, 'Business'),
(5, 'Science');

INSERT INTO Authors (author_id, first_name, last_name, bio) VALUES
(1, 'Robert C.', 'Martin', 'Software engineering author known for Clean Code.'),
(2, 'Andrew', 'Hunt', 'Software developer and co-author of The Pragmatic Programmer.'),
(3, 'David', 'Thomas', 'Software developer and co-author of The Pragmatic Programmer.'),
(4, 'George', 'Orwell', 'English novelist and essayist.'),
(5, 'Yuval Noah', 'Harari', 'Historian and author of Sapiens.'),
(6, 'Jane', 'Austen', 'English novelist.'),
(7, 'Andrew S.', 'Tanenbaum', 'Computer scientist and textbook author.'),
```

```
(8, 'Frederick P.', 'Brooks Jr.', 'Computer scientist and software project management author.'),
(9, 'Stephen', 'Hawking', 'Theoretical physicist and science author.');
```

3. Insert books and author mappings

```
INSERT INTO Books (book_id, title, isbn, category_id, publication_year, total_copies, available_copies) VALUES
(1, 'Clean Code', '9780132350884', 1, 2008, 5, 3),
(2, 'The Pragmatic Programmer', '9780201616224', 1, 1999, 4, 3),
(3, 'Nineteen Eighty-Four', '9780451524935', 2, 1949, 6, 5),
(4, 'Sapiens', '9780062316097', 3, 2011, 4, 4),
(5, 'Pride and Prejudice', '9780141439518', 2, 1813, 3, 2),
(6, 'Computer Networks', '9780132126953', 1, 2012, 5, 4),
(7, 'The Mythical Man-Month', '9780201835953', 4, 1975, 3, 3),
(8, 'A Brief History of Time', '9780553380163', 5, 1988, 4, 4);

INSERT INTO Book_Authors (book_id, author_id) VALUES
(1, 1),
(2, 2), (2, 3),
(3, 4),
```

```
(4, 5),
(5, 6),
(6, 7),
(7, 8),
(8, 9);
```

4. Insert members and loan transactions

```
INSERT INTO Members (member_id, first_name, last_name, email, phone, membership_date, status) VALUES
(1, 'Asha', 'Rao', 'asha.rao@example.edu', '+91-9000000001', '2026-01-10', 'Active'),
(2, 'Rahul', 'Menon', 'rahul.menon@example.edu', '+91-9000000002', '2026-02-02', 'Active'),
(3, 'Priya', 'Sharma', 'priya.sharma@example.edu', '+91-9000000003', '2026-02-15', 'Active'),
(4, 'Karthik', 'Iyer', 'karthik.iyer@example.edu', '+91-9000000004', '2026-03-01', 'Active'),
(5, 'Meera', 'Nair', 'meera.nair@example.edu', '+91-9000000005', '2026-03-18', 'Active'),
(6, 'Vikram', 'Das', 'vikram.das@example.edu', '+91-9000000006', '2026-04-05', 'Suspended');

INSERT INTO Loans (loan_id, book_id, member_id, loan_date, due_date, return_date, status) VALUES
(1, 1, 1, '2026-10-01', '2026-10-15', NULL, 'Overdue'),
(2, 1, 2, '2026-10-20', '2026-11-03', NULL, 'Issued'),
(3, 2, 3, '2026-10-05', '2026-10-19', NULL, 'Overdue'),
(4, 3, 4, '2026-10-10', '2026-10-24', NULL, 'Issued'),
(5, 5, 5, '2026-10-21', '2026-11-04', NULL, 'Issued'),
```

```
(6, 6, 1, '2026-10-22', '2026-11-05', NULL, 'Issued'),
(7, 1, 3, '2026-08-01', '2026-08-15', '2026-08-12', 'Returned'),
(8, 2, 4, '2026-07-02', '2026-07-16', '2026-07-14', 'Returned'),
(9, 3, 1, '2026-06-10', '2026-06-24', '2026-06-20', 'Returned'),
(10, 4, 2, '2026-09-05', '2026-09-19', '2026-09-15', 'Returned'),
(11, 5, 3, '2026-09-08', '2026-09-22', '2026-09-21', 'Returned'),
(12, 7, 6, '2026-08-20', '2026-09-03', '2026-09-01', 'Returned'),
(13, 8, 5, '2026-07-15', '2026-07-29', '2026-07-25', 'Returned'),
(14, 6, 4, '2026-09-12', '2026-09-26', '2026-09-23', 'Returned');
```

5. Verify the number of inserted records

```
SELECT 'Categories' AS table_name, COUNT(*) AS row_count FROM Categories
UNION ALL SELECT 'Authors', COUNT(*) FROM Authors
UNION ALL SELECT 'Books', COUNT(*) FROM Books
UNION ALL SELECT 'Book_Authors', COUNT(*) FROM Book_Authors
UNION ALL SELECT 'Members', COUNT(*) FROM Members
UNION ALL SELECT 'Loans', COUNT(*) FROM Loans;
```

table_name	row_count
Categories	5
Authors	9
Books	8
Book_Authors	9
Members	6
Loans	14

6. Data consistency notes

- Each Books.category_id exists in Categories; each Book_Authors row references an existing book and author.
- Each Loans.book_id and Loans.member_id refers to an existing book and member.
- There are six non-returned loans (status Issued or Overdue). The sample available_copies values are set consistently with these outstanding loan counts for each title.
- The two overdue records have past due dates relative to the dataset date, 23 October 2026. Sample dates are fixed to make the reports repeatable; they do not auto-roll forward.

The complete setup plus inserts is available in LibSphere_Full_Setup_and_Queries.sql. Run the DDL before inserts on a fresh database; rerunning the insert section against existing IDs will produce duplicate-key errors unless the database is recreated or cleared.